

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

**Duration : 45 Minutes
Total Marks:20**

Annexure A

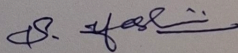
Descriptive Written Test

Code of Conduct:

I Hashini S/192524035 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



3/08/26

Hashini S
192524035

Describe Test

CSA1703 - Artificial Intelligence

16
20

1. Introduction:

A Smart home Controls the Lightening, temperature and sensors such as motion detectors, temperature sensors and other input data. It learns user preferences to provide Comfort and Security.

Relating to Intelligent Agents:

1. Simple Reflex Agent:

- It acts only on the current inputs with using the predefined rules. It does not know about previous state.

Working in Smart Home

⇒ if motion is detected - TURN ON Lights

⇒ if no motion - TURN OFF Lights

⇒ if the temperature is above 30°C - TURN ON AC.

Advantages:

- Fast Response
- Easy to implement
- Low Computational cost.

Disadvantages:

- May make incorrect decisions in the complex situations
- No memory.

2. Model-Based Agent

- It maintains an internal model of the environment. It also remembers the previous activities.

Working:

- Remembers whether someone is already in the room.
- Keeps tracking the room occupancy
- Adjusting the lights with previous sensor data.

Advantages:

- Better decision making
- Works even with incomplete sensor data.

Disadvantages:

- Requires memory
- More complex to implement.

Goal-Based Agent.

- It selects actions that help achieve a specific goal.

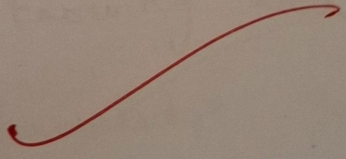
Working :

- If everyone leaves the house
- Lock doors
- Turn off lights
- Switch AC to energy saving mode

Advantages :

- Plans actions intelligently
- Flexible in different situations.

Disadvantages :

- Slower due to planning
 - Needs search algorithms.
- 

4. Utility-Based Agent:

- It chooses the Actions that provide maximum utility benefits.

Working:

- Saves
- Maintains 24°C instead of 18°C to save the electricity while keeping users comfortable.

Advantages:

- Optimizes Energy Efficiency.
- Learns user preferences.

Disadvantages:

- Complex Utility Calculations
- Higher Computational Task.

Best choice:

- Hybrid Agent [Model-Based + Goal-Based + Utility Based] is more effective.

Reason:

- Model-based remembers the Previous States
- Goal-based achieves comfort
- Utility-based optimizes energy efficiency.

Conclusion:

A hybrid intelligent agent provides the best balance between comfort, energy efficiency and security.

2. Introduction:

A Robot Placed in Unknown Maze must find the exit without knowing the maze layout. Since it has no prior knowledge, it uses Uninformed Search algorithms.

1. Uniform Cost Search (UCS)
- It expands the path with
lowest total cost first.

Working:

Start from initial position



Calculate total path cost



Select least-cost path



Stops when the exit reached.

Advantages:

- Finds the optimal path
- Works well with different path costs.

Disadvantages:

- High memory usage
- Slow for large mazes.

Time Complexity: Exponential (depends on branching factors and cost).

Space Complexity: Exponential.

Interactive Deepening Search (IDS)

- IDS repeatedly performs Depth First Search with increasing depth limits.

Working:

- Search depth = 0
- Then depth = 1
- Then depth = 2
- Continue until exit found.

Advantages:

- Requires very little memory
- Finds shallow solutions.

Disadvantages:

- Repeats Searches
- May take more time.

Complexity:

- Time: $O(b^d)$
- Space: $O(b \times d)$

Relation to Intelligent Agents

1. Simple Reflex Agent

- Reacts only to current state
- Cannot plan a path

2. Model-Based Agent

- Remember previous state
- Avoids revisiting
- Better than simple reflex.

3. Goal-Based Agent

- Goal is to reach maze exit
- Uses search algorithm like UCS or IDS
- plans before moving.

4. Utility-Based Agent

- chooses the best action

- Time
- Energy
- Safety

- Useful when multiple exists | path costs

exists

Combination:

Goal - Based Agent + Uniform Cost Search

Justification:

- Goal-based focuses on reaching end and achieves comfort.
- UCS guarantees the least-cost path
- Suitable when movement costs vary.

If memory limited, then:-

- Goal based + Interactive Deeping Search
- It is better choice because IDS requires much less memory.

Conclusion:

- UCS is best for finding optimal path.
- IDS is best when memory is limited.
- Goal - Based Agent Combined with UCS which provides effective and intelligent maze navigation.